

# Vivek Chavan | PhD Candidate at TU Berlin and Researcher at Fraunhofer IPK

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**SUMMARY:** Computer Vision and Robotics researcher focused on industrial ML systems. Research on *Multi-modal Perception, Continual Learning, Data Curation, Vision-Language(-Action) models, Embodied Agents*.

## WORK EXPERIENCE

### Research Associate | Fraunhofer IPK, Berlin

Nov 2022 - present

Led and contributed to research and development projects; established research collaboration with Meta Reality Labs and other institutions; developed and deployed novel multimodal ML solutions for classification, segmentation, quality assessment, and MLOps; spearheaded international research consortia and coordinated the conception and application process for €3M+ in research funding (€1.8M successfully awarded as of Q2 2026).

### Intern and Student Assistant | Fraunhofer IPK, Berlin

Oct 2021 - Oct 2022

Evaluated multimodal foundation models and deep learning approaches for production applications. Curated two datasets and resolved data quality bottlenecks in industrial Computer Vision.

### CAE Engineer | Tata Technologies & Jaguar Land Rover, India & UK

July 2014 – Aug 2019

Developed and validated large-scale CFD and CAE simulations to optimise vehicle performance; co-authored two technical papers introducing new modelling methodologies; collaborated with cross-functional engineering teams across design, testing, and integration; represented the team in international collaborations.

## EDUCATION

2023 - Present PhD at *Technical University of Berlin*.

Continual Egocentric/Exocentric Perception for Procedural Understanding & Embodied Action

2019 - 2022 Master's Degree at *RWTH Aachen University*.

Thesis: Green Incremental Learning.

[Thesis Grade: 1,0 (Perfect Score)]

2010 - 2014 Bachelor's Degree at *Pune University, India*.

[Graduated Top of the Class]

## SELECTED FIRST AUTHOR PUBLICATIONS

[GOOGLE SCHOLAR](#)

V. Chavan, Y. Imgrund, T. Dao, et al., "IndEgo: A Dataset of Industrial Scenarios and Collaborative Work for Egocentric Assistants." *Advances in Neural Information Processing Systems (NeurIPS 2025)*.

[Established benchmarks and baselines for Mistake Detection, Video Question Answering, collaborative work and task summarization in human-centric industrial settings.]

V. Chavan, J. Krüger, "AI-based worker guidance in assembly and disassembly operations using multimodal ego/exo-centric data capture and structured task knowledge." *CIRP Annals - Manufacturing Technology, 2026*.

[Encoded temporal multimodal information from egocentric expert demonstrations into structured representations. Enabled context-aware worker guidance and procedural documentation.]

V. Chavan, P. Koch, M. Schlüter, et al., "Towards Realistic Evaluation of Industrial Continual Learning Scenarios with an Emphasis on Energy Consumption and Computational Footprint." *Proc. IEEE/CVF International Conference on Computer Vision (ICCV 2023)*.

[Established evaluation protocols for incremental learning by integrating energy consumption and computational load as primary metrics for real-world adoption.]

V. Chavan, O. Heimann, J. Krüger, "On the Application of Egocentric Computer Vision to Industrial Inspection." *Proc. European Conference on Computer Vision Workshops (ECCVW 2024)*.

[Fused narration, gaze, and hand-pose for automated annotation in industrial inspection via a multimodal pipeline. Benchmarked egocentric perception and quantified ego-to-exocentric domain shifts.]

V. Chavan, O. Heimann, J. Krüger, "A System 1 and System 2 Perspective on Continual Learning for Practical Implementation." *Proc. European Conference on Computer Vision Workshops (ECCVW 2024)*.

[Integrated contextual semantic encoding with robust pretraining to enhance model generalization and mitigate catastrophic forgetting across varied data distributions.]

V. Chavan, P. Koch, M. Schlüter, et al., “Active Data Collection and Management for Real-World Continual Learning via Pretrained Oracle.” *Proc. IEEE/CVF Computer Vision and Pattern Recognition Workshops (CVPRW 2024)*.

[Developed a framework for active coreset selection and distribution matching in non-curated streams using a pretrained oracle. Demonstrated improvements over state-of-the-art exemplar selection strategies.]

V. Chavan, Y. Zhang, J. Krüger, “PROST: Procedural State-Transition Modeling for Error Detection in Egocentric and Exocentric Data.” *Currently in preparation, targeting ICLR 2027*.

[A procedural task modeling approach that represents tasks as state-transition systems grounded in multimodal egocentric and exocentric video data; for error recognition in single and multi-agent settings.]

V. Chavan, J. Krüger, “A Framework for Egocentric and Exocentric Procedural Understanding via Temporal Segmentation and Semantic Abstraction.” *Currently in preparation, targeting WACV 2027*.

[Converting multimodal streams into structured event-based memory representations by segmenting continuous experience, abstracting key interactions, and selectively retaining information for efficient downstream use.]

## OPEN-SOURCE RESEARCH DATASETS

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*IndEgo [2025]*: A large-scale multimodal ego/exo dataset capturing RGB, time-synchronized gaze, motion data in procedural tasks and diverse industrial use-cases (e.g., Assembly/disassembly, logistics, repair).

*MVIP [2024]*: Multimodal RGB-D benchmark of 300+ industrial parts across 10 calibrated views. Includes semantic masks, physical properties, and text metadata; utilized for modality fusion and few-shot transfer.

*InVar-100 [2023]*: Industrial object recognition benchmark containing 100 distinct object classes across varied environmental contexts. Simulates the distribution shift and visual noise characteristic of industrial deployments.

## TECHNICAL REPORTS, POSITION PAPERS & WHITE PAPERS

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V. Chavan, A. Cenaj, S. Shen, et al., “Feeling Machines: Ethics, Culture, and the Rise of Emotional AI”. *Technical Report, arXiv:2506.12437*, 2025.

[Led an interdisciplinary analysis of the ethical and cultural risks of AI assistants and agents. Developed ten strategic recommendations for regulatory safeguards and technical transparency.]

V. Chavan, et al., “Learning-Based Decision Systems for Circular Production.” *Forthcoming, 2026*.

[Proposed an ML-based framework for decision-making and automation in (re)manufacturing processes. Exploring targeted data curation, multimodal data fusion, and predictive modeling approaches.]

## ACADEMIC SERVICE AND LEADERSHIP

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*Reviewer and Sub-reviewer*: CVPR 2026, ECCV 2026, and NeurIPS 2026 (invited), and related workshops.

*Mentorship*: Supervised 5 Bachelor/Master theses, 9 research internships, and 9 student semester project participants at Fraunhofer IPK/TU Berlin.

*AI Grid Representative (Berlin Area)*: Coordinating interdisciplinary research, networking, and knowledge exchange for the German national AI initiative.

## CERTIFICATIONS

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NVIDIA Multimodal AI Agents & RAG Pipelines for Production; Coursera MLOps Specialization; TensorFlow Developer Certificate; Coursera Deep Learning & TensorFlow Specializations; Lean Six Sigma Yellow Belt.

## SKILLS & INTERESTS

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*Technical*: PyTorch, TensorFlow, Python, Multimodal Foundation Models, Continual Learning, Robotics (ROS), MLOps, Linux, Git, Large-scale CAE/CFD Simulations.

*Interests*: Scientific Writing, Public Speaking, Fitness and Wellbeing.

Vivek Chavan